EnterpriseOS

Business Requirements Document (BRD) + Software Requirements Specification (SRS)

Phase 3: Commercial, Financial and Asset Operations

Document control	Value
Version	1.0 - Client review draft
Status	Proposed; client confirmation required
Project	EnterpriseOS - multi-tenant SaaS ERP
Prepared for	Client / Business Sponsor
Stack	React + TypeScript; Java Spring Boot; MySQL 8.4 LTS
Prerequisite	Approved and delivered Phase 2 baseline
Date	11 November 2026
Approval	Pending client sign-off

Purpose: Determine client decisions, specify the proposed phase scope and provide verifiable approval and acceptance gates. Any default described here is a proposal, not a confirmed client requirement.

1. Executive Summary & Business Outcomes

Extend the platform with CRM, sales, inventory/procurement, expense management, invoicing and payroll integrations/calculation policies after Phase 2 workforce data is accepted.

Phase 3 consumes the shared identity, tenancy and role model defined in Phase 1. Functional changes are implemented as new bounded modules in the same Spring Boot modular monolith. Existing security controls and audit records remain mandatory.

2. Prerequisites, Dependencies & Stakeholders

Dependency	Entry requirement
Prior phase	Phase 2 released, accepted and regression-tested.
Tenant & identity	Active tenant context, membership validation, session security and permission model available.
Reference data	Departments, staff, currencies, policies and other dependent master data approved as applicable.
Stakeholders	Client sponsor, product owner, department SMEs, information security contact, UAT owner nominated.
Delivery assets	Source repository, staging environment, MySQL migration process, CI tests, email service available.

Responsible stakeholders: Sales Manager, Sales Representative, Procurement Officer, Warehouse Manager, Finance Manager, Payroll Officer, Employee Approver, Tenant Auditor.

3. Client Discovery - Required Inputs Before Scope Freeze

1. Which country/state laws, tax rules, invoicing requirements and payroll compliance rules apply?
   Client answer / reference: _____

2. What currencies, exchange rates, fiscal year and tax rounding rules are approved?
   Client answer / reference: _____

3. Does the client need accounting ledger or only invoicing/expense tracking?
   Client answer / reference: _____

4. What inventory valuation model applies (weighted-average/FIFO) and can negative stock occur?
   Client answer / reference: _____

5. Are goods receipts partial, returnable and subject to inspection?
   Client answer / reference: _____

6. Which sales pipeline stages, quote rules, approval thresholds and tax categories apply?
   Client answer / reference: _____

7. Which payroll earnings, deductions, payment cycles and bank formats are required?
   Client answer / reference: _____

8. Who owns final legal review of payslips, tax calculations and invoice templates?
   Client answer / reference: _____

4. Functional Requirements - Traceability Baseline

4.1 CRM & Sales Operations

- P3-FR-0101 | Manage leads, accounts, contacts and lead ownership
- P3-FR-0102 | Configure pipeline stages and move opportunities with stage history
- P3-FR-0103 | Record activities, follow-up reminders, notes and customer attachments
- P3-FR-0104 | Generate quotes with line items, taxes and expiration
- P3-FR-0105 | Convert approved quotes to orders based on client process

4.2 Inventory & Procurement

- P3-FR-0201 | Maintain product/item master, categories, units and warehouse locations
- P3-FR-0202 | Record stock receipts, transfers, issues and adjustments as immutable movements
- P3-FR-0203 | Implement supplier master and purchase requisitions/orders
- P3-FR-0204 | Match goods receipts to purchase orders with quantity tolerance rules
- P3-FR-0205 | Monitor reorder points, stock valuation policy and movement reports

4.3 Expense, Invoice & Receivables

- P3-FR-0301 | Capture staff expense claims with receipts and categories
- P3-FR-0302 | Approve expenses against budgets, thresholds and cost centers
- P3-FR-0303 | Create invoices, line items, tax breakdowns and credit notes
- P3-FR-0304 | Record customer payments and balance reconciliation statuses
- P3-FR-0305 | Export finance-ready reports and tax records based on approved jurisdiction

4.4 Payroll & Compensation

- P3-FR-0401 | Configure payroll periods, salary components and effective-dated employee compensation
- P3-FR-0402 | Import approved attendance/leave inputs and support deductions as configurable rules
- P3-FR-0403 | Run draft payroll with review, lock, versioning and re-run safeguards
- P3-FR-0404 | Generate payslips and role-limited payroll register
- P3-FR-0405 | Export approved bank transfer file with agreed format and controls

P0 = mandatory only after signed baseline. P1 = optional enhancement subject to separate client approval. Details such as calculation formulas, limits and country-specific rules must be signed off with worked examples.

5. Business Rules, Validation & Data Integrity

- BR-3-01: Tenant and branch ownership must be validated for every CRM, warehouse, purchase and finance resource.
- BR-3-02: Posted stock movement is append-only; corrections create compensating movements, not silent edits.
- BR-3-03: Issue or transfer stock must respect stock availability unless approved negative-stock policy applies.
- BR-3-04: Posted invoices and finalized payroll runs cannot be silently changed; use credit/reversal/version processes.
- BR-3-05: Money values use fixed-precision decimal fields and approved rounding/tax allocation conventions.
- BR-3-06: Payroll must be reviewed by authorized maker/checker roles before finalization.
- BR-3-07: No transfer of actual funds is performed; a bank export is only a file and must be verified manually.
- BR-3-08: No statutory tax or payroll compliance is claimed until localized rules are specified, reviewed and validated.

6. Roles, Permissions & Segregation of Duties

Client must approve a role-permission matrix at action level (view, create, edit, approve, export, delete). Default-deny access is enforced by Spring Security and service-layer object authorization. UI restrictions never substitute for API enforcement.

Role	Primary scope (proposed)
Sales Manager	Access to assigned module/actions only; final privileges require client-signed matrix.
Sales Representative	Access to assigned module/actions only; final privileges require client-signed matrix.
Procurement Officer	Access to assigned module/actions only; final privileges require client-signed matrix.
Warehouse Manager	Access to assigned module/actions only; final privileges require client-signed matrix.
Finance Manager	Access to assigned module/actions only; final privileges require client-signed matrix.
Payroll Officer	Access to assigned module/actions only; final privileges require client-signed matrix.
Employee Approver	Access to assigned module/actions only; final privileges require client-signed matrix.
Tenant Auditor	Access to assigned module/actions only; final privileges require client-signed matrix.

Control requirements: no unapproved self-approval, separation of preparer/approver where mandated, auditing of privileged changes, and tested cross-tenant isolation for all tenant-owned tables.

7. Business Workflows & User Journeys

- Flow 1: Sales rep qualifies lead -> opens opportunity -> prepares quote -> manager approval -> customer acceptance -> invoicing.
- Flow 2: Requester raises purchase request -> approval -> PO issued -> warehouse receives goods -> stock movement updates.

- Flow 3: Employee submits receipt expense -> manager approval -> finance review -> recorded as reimbursable/exportable.
- Flow 4: Payroll admin prepares period -> approved attendance sync -> draft calculations -> finance review -> lock -> payslips/export.

8. UI Screens & User Experience

Screen group	Minimum proposed interface
CRM & Sales Operations	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Inventory & Procurement	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Expense, Invoice & Receivables	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states
Payroll & Compensation	List/search/filter with pagination; create/edit/detail forms; permission-aware actions; review, error, empty and audit states

Common UX: accessible input labels, confirmation for irreversible actions, reliable error feedback, mobile-responsive layout, keyboard navigation, loading indicators, localization/currency/timezone where applicable. Wireframes require client approval.

9. Data Model & MySQL Entities

Domain module	Proposed core tables (tenant-scoped unless global)
CRM & Sales Operations	crm_leads, crm_accounts, crm_contacts, opportunities, sales_activities, quotes, quote_lines
Inventory & Procurement	items, warehouses, stock_movements, suppliers, purchase_requests, purchase_orders, goods_receipts
Expense, Invoice & Receivables	expense_claims, expense_lines, cost_centers, invoices, invoice_lines, credit_notes, payment_records
Payroll & Compensation	salary_components, employee_compensation, payroll_runs, payroll_run_items, payroll_adjustments, payslips

Schema principles: InnoDB foreign keys and appropriate unique constraints; tenant_id on tenant-owned business rows, UTC event timestamps, decimal types for money, effective-date history where necessary, optimistic versioning/locking for concurrent state changes, indexes based on measured queries, Flyway/Liquibase migrations. Do not silently cascade-delete finance, HR or audit evidence.

Client data decisions: retention, import source format, data stewardship and field-level privacy must be confirmed before database design approval. A detailed ERD, dictionary and migration/backfill plan are deliverables, not presumed existing artifacts.

10. REST API Contract - Illustrative Scope

Method	Proposed path	Access
GET/POST	/api/v1/org/crm/leads	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/crm/opportunities	Verified tenant role; platform-only

		where /platform
POST	/api/v1/org/crm/quotes	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/inventory/items	Verified tenant role; platform-only where /platform
POST	/api/v1/org/inventory/movements	Verified tenant role; platform-only where /platform
POST	/api/v1/org/procurement/orders	Verified tenant role; platform-only where /platform
POST	/api/v1/org/procurement/receipts	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/expenses	Verified tenant role; platform-only where /platform
GET/POST	/api/v1/org/invoices	Verified tenant role; platform-only where /platform
POST	/api/v1/org/payroll/runs	Verified tenant role; platform-only where /platform
POST	/api/v1/org/payroll/runs/{id}/finalize	Verified tenant role; platform-only where /platform

API standards: version /api/v1, DTO validation, consistent error envelope and correlation IDs, OpenAPI documentation, stable pagination/filters, idempotency for critical operations, conflict handling (409), throttling for sensitive endpoints and audit hooks.

11. Non-Functional & Security Requirements

Category	Acceptance baseline / to confirm
Availability	Client to specify hosting/SLA, maintenance window and support hours; no implied uptime warranty.
Performance	Confirm p95 read/write latency and concurrent-user workload using realistic records and tenant distribution.
Security	TLS, authorization on every operation, secure secrets, validated uploads/exports, no password/token logging.
Privacy	Personal and financial data minimization; retention, consent, export and deletion policy signed by client.
Reliability	Backups and restore drill, concurrency testing, idempotent events, transactional state changes.
Accessibility	Keyboard-operable WCAG 2.2 AA target where explicitly agreed; browser matrix for Chrome/Edge/Firefox.
Quality	Unit, integration, API-contract, cross-tenant, regression and E2E tests in CI; security review gates.
Observability	Structured logs, metrics, health checks, alerts and audit trail without sensitive values.

12. In Scope / Explicit Exclusions

Included in Phase 3	Deferred or excluded
CRM & Sales Operations: agreed requirements	Direct payment gateway charging or bank transfer execution
Inventory & Procurement: agreed requirements	Full double-entry accounting/ERP ledger unless separately scoped
Expense, Invoice & Receivables: agreed requirements	Point-of-sale and e-commerce storefront

Payroll & Compensation: agreed requirements	Multi-country payroll compliance without country-specific specification
CRM & Sales Operations: agreed requirements	External ERP, tax filing or banking integrations without signed API contracts

All third-party services, subscriptions, tax/legal assurance, external integrations and migration volumes require written scope and cost approval. AI, ML and generative AI features are explicitly out of scope for EnterpriseOS.

13. Acceptance Criteria & User Acceptance Testing (UAT)

- UAT-P3-01: Lead is assigned within tenant and stage history is preserved.
- UAT-P3-02: Stock quantity and valuation reconcile after goods receipt, issue, return and transfer test cases.
- UAT-P3-03: Concurrent stock issues cannot create prohibited negative stock.
- UAT-P3-04: Approved quote-to-invoice conversion preserves line items and amounts.
- UAT-P3-05: Invoice totals and tax/rounding rules match signed calculation examples.
- UAT-P3-06: Employee cannot access another employee payroll detail without approved role.
- UAT-P3-07: Finalized payroll run is immutable; correction follows approved reversal process.
- UAT-P3-08: Non-authorized users cannot finalize expenses, invoices or payroll.

Client approves UAT evidence with expected result, actual result, test data, screenshot/log reference, tester, date and disposition. Critical/high defects must be fixed or explicitly accepted with documented risk before production release.

14. Development Deliverables & Release Gates

Stage	Deliverables	Exit gate
Discovery	Approved BRD/SRS, policy examples, role-permission matrix, dependencies	Requirements sign-off
Design	Wireframes, architecture notes, ERD/data dictionary, API spec, threat/edge-case review	Design sign-off
Build	React pages, Spring Boot modules, MySQL migrations, tests, audit hooks	Functional demonstration
Verify	Automated CI, security/regression tests, data migration checks, UAT runs	UAT approval
Release	Deployment, rollback/backup procedure, API docs, release notes, handover/training	Go-live authorization

Commercial schedule, milestones and cost remain TBD. They depend on approved scope, policy complexity, third-party licensing, existing-data quality, staff capacity and client review cycles. Changes after sign-off use a documented change request with cost and schedule impact.

15. Assumptions, Risks & Open Decisions

Topic	Risk / resolution
Prior phase dependency	Untested upstream permissions or data structures can invalidate this phase; require regression gate.
Policy ambiguity	Unapproved rule formulas or approval paths cause rework; require signed examples.
Tenant privacy	Cross-tenant and overprivileged exports are release blockers;

	negative tests required.
Data migration	Legacy fields may be incomplete/duplicated; profile and pilot migration before bulk import.
Performance targets	Capacity and realistic datasets are unknown; benchmark after workload approval.
Compliance & legal	Client owns jurisdictional requirements; engineering does not certify legal compliance.
Scope changes	Use issue/decision log and signed change order before adding new integrations/modules.

16. Client Decisions & Requirements Checklist

☐ Which country/state laws, tax rules, invoicing requirements and payroll compliance rules apply? Answer/reference: _____

☐ What currencies, exchange rates, fiscal year and tax rounding rules are approved? Answer/reference: _____

☐ Does the client need accounting ledger or only invoicing/expense tracking? Answer/reference: _____

☐ What inventory valuation model applies (weighted-average/FIFO) and can negative stock occur? Answer/reference: _____

☐ Are goods receipts partial, returnable and subject to inspection? Answer/reference: _____

☐ Which sales pipeline stages, quote rules, approval thresholds and tax categories apply? Answer/reference: _____

☐ Approved roles/permission matrix attached

☐ Wireframes, report examples and validation rules approved

☐ Hosting/security/retention/deployment requirements agreed

☐ UAT cases and authorized signatories agreed

17. Approval & Sign-off

Signatures indicate approval of the stated Phase 3 requirements baseline and acceptance conditions only. This document does not establish pricing, implementation warranties, or regulatory certifications.

Role	Name / signature	Date
Client Sponsor	_____	_____
Client Product Owner	_____	_____
Technical Delivery Lead	_____	_____

End of Phase 3 requirements document | Version 1.0 | Status: Client review draft